MACHINE LEARNING ASSIGNMENT 5

1. R-squared or Residual Sum of Squares (RSS) which one of these two is a better measure of goodness of fit model in regression and why?

R-squared is a statistical measure that represents the proportion of the variance in the dependent variable that is explained by the independent variables in the regression model.

It ranges from 0 to 1, with 1 indicating a perfect fit and 0 indicating no relationship between the independent and dependent variables.

Residual sum of squares (RSS) is another important metric used to evaluate the goodness of fit of a regression model.

It measures the sum of the squared differences between the predicted values and the actual values of the dependent variable. The smaller the RSS, the better the fit of the model. RSS is useful for identifying specific areas where the model may not be performing well, such as outliers or regions with high variance.

Hence Residual Sum of Squares (RSS) is a better measure of goodness of fit model in regression.

2. What are TSS (Total Sum of Squares), ESS (Explained Sum of Squares) and RSS (Residual Sum of Squares) in regression. Also mention the equation relating these three metrics with each other.

TSS (Total Sum of Squares) represents the total variability in the response variable (y), and is calculated as the sum of the squared differences between each observation of y and the mean of y.

ESS (Explained Sum of Squares) represents the variability in y that is explained by the regression model, and is calculated as the sum of the squared differences between the predicted values of y ($\hat{y}$) and the mean of y.

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total sum of squares (TSS) = explained sum of squares (ESS) + residual sum of squares (RSS).

3. What is the need of regularization in machine learning?

Need of regularization in machine learning to properly fit the model to our test set. Regularization techniques help reduce the possibility of overfitting and help us obtain an optimal model.

4. What is Gini–impurity index?

Gini Impurity is a measurement used to build Decision Trees to determine how the features of a dataset should split nodes to form the tree. More precisely, the Gini Impurity of a dataset is a number between 0-0.5, which indicates the likelihood of new, random data being misclassified if it were given a random class label according to the class distribution in the dataset.

5. Are unregularized decision-trees prone to overfitting? If yes, why?

Yes, unregularized decision trees are prone to overfitting because they can create a complex model that fits the training data too well, including the noise and outliers in the data. Overfitting occurs when the model captures the behaviour of the training data so closely that it fails to generalize well to new, unseen data.

6. What is an ensemble technique in machine learning?

Ensemble methods are techniques that aim at improving the accuracy of results in models by combining multiple models instead of using a single model. The combined models increase the accuracy of the results significantly. This has boosted the popularity of ensemble methods in machine learning.

7. What is the difference between Bagging and Boosting techniques?

Bagging
It increases the accuracy of models through decision trees, which reduces variance to a large extent. The reduction of variance increases accuracy, eliminating overfitting, which is a challenge to many predictive models.

Boosting

Boosting is an ensemble technique that learns from previous predictor mistakes to make better predictions in the future. The technique combines several weak base learners to form one strong learner, thus significantly improving the predictability of models. Boosting works by arranging weak learners in a sequence, such that weak learners learn from the next learner in the sequence to create better predictive models.

8. What is out-of-bag error in random forests?

The out-of-bag (OOB) error is the average error for each calculated using predictions from the trees that do not contain in their respective bootstrap sample. This allows the RandomForestClassifier to be fit and validated whilst being trained.

9. What is K-fold cross-validation?

It is a *data partitioning strategy* so that it can effectively use your dataset to build *a more generalized model*.
The main intention of doing any kind of machine learning is to develop a more generalized model which can perform well on *unseen data*.
One can build a perfect model on the training data with 100% accuracy or 0 error, but it may fail to generalize for unseen data. So, it is not a good model. It overfits the training data.
Machine Learning is all about *generalization* meaning that model's performance can only be measured with data points that have never been used during the training process.

10. What is hyper parameter tuning in machine learning and why it is done?

Hyperparameters are the settings that can be tuned before running a training job to control the behavior of an ML algorithm.
They can have a big impact on model training as it relates to training time, infrastructure resource requirements (and as a result cost), model convergence and model accuracy.

11. What issues can occur if we have a large learning rate in Gradient Descent?

if we have a large learning rate, the algorithm may bypass the local minimum and overshoot ,in order for Gradient Descent to work, we must set the learning rate to an appropriate value.

This parameter determines how fast or slow we will move towards the optimal weights. If the learning rate is very large we will skip the optimal solution. If it is too small we will need too many iterations to converge to the best values. So using a good learning rate is crucial.

12. Can we use Logistic Regression for classification of Non-Linear Data? If not, why?

No, logistic regression can not use for classification of Non-Linear Data.
However, if the data is not linearly separable, logistic regression may not perform well, and other methods such as decision trees, support vector machines, or neural networks may be more appropriate.

One way to use logistic regression for non-linear data is to transform the features of the data using non-linear functions, such as polynomial functions or sigmoidal functions. This approach is known as feature engineering, and it allows the model to capture non-linear relationships between the features and the target variable.

Another approach to handle non-linear data is to use a variant of logistic regression called "non-linear logistic regression" or "non-linear probabilistic classification." This model uses a non-linear function, such as a sigmoid to map the linear combination of the input features to the output probabilities.

13 Differentiate between Adaboost and Gradient Boosting.

- AdaBoost is the first designed boosting algorithm with a particular loss function. On the other hand, Gradient Boosting is a generic algorithm that assists in searching the approximate solutions to the additive modelling problem. This makes Gradient Boosting more flexible than AdaBoost.

- AdaBoost minimises loss function related to any classification error and is best used with weak learners. The method was mainly designed for binary classification problems and can be utilised to boost the performance of decision trees.

- The technique of Boosting uses various loss functions. In case of Adaptive Boosting or AdaBoost, it minimises the exponential loss function that can make the algorithm sensitive to the outliers.

14. What is bias-variance trade off in machine learning?

The process of finding the right balance between the bias and variance of the model is called the Bias-Variance trade-off. It is basically a way to make sure the model is neither overfitted or underfitted in any case.

15. Give short description each of Linear, RBF, Polynomial kernels used in SVM.

Linear used in SVM:

Linear SVM is used for linearly separable data, which means if a dataset can be classified into two classes by using a single straight line, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.

RBF (radial basis function) used in SVM:

RBF Kernel is popular because of its similarity to K-Nearest Neighborhood Algorithm. It has the advantages of K-NN and overcomes the space complexity problem as RBF Kernel Support Vector Machines just needs to store the support vectors during training and not the entire dataset.

Polynomial kernels used in SVM

A polynomial kernel is a kind of SVM kernel that uses a polynomial function to map the data into a higher-dimensional space and allowing learning of non-linear models.